LCA Module v3 — Merge Specification Document

Version: 3.0.0

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1. Basis and Rationale for the Merge

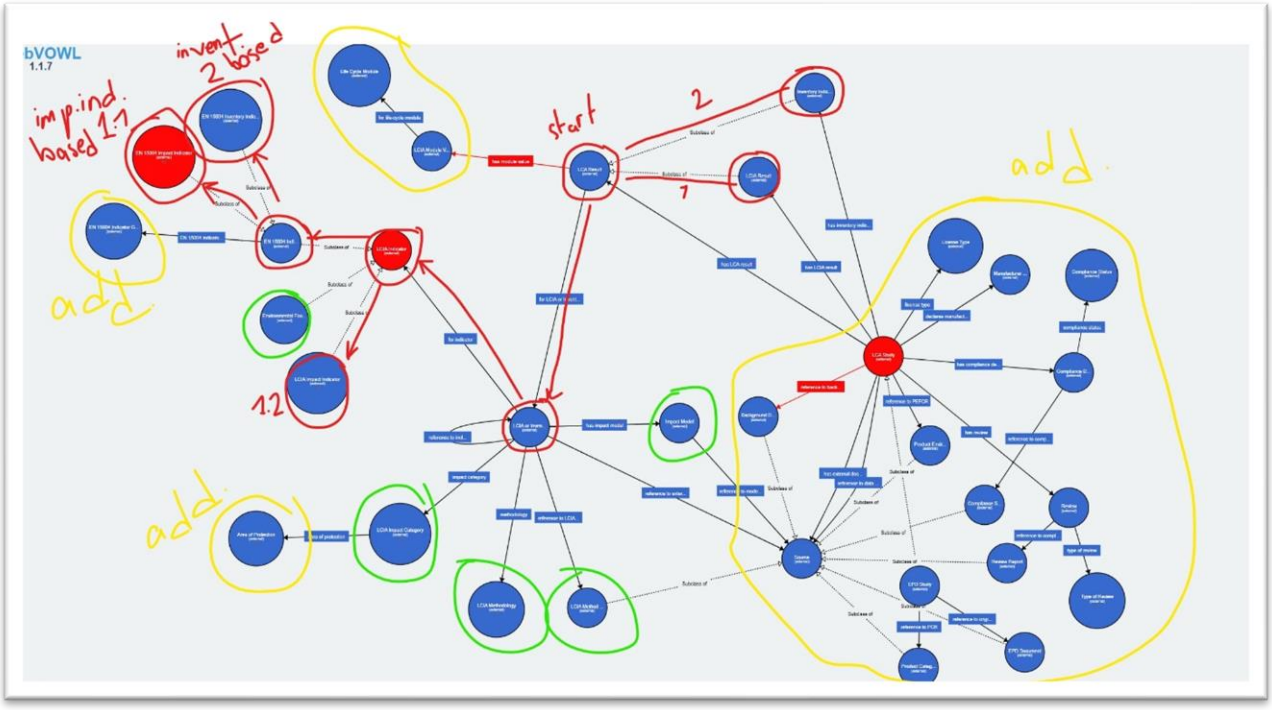
The LCA module v3 results from the integration of two independently developed v2.0:

- **Maki v2.0** — developed by Maki, structured around ILCD+EPD with a lean manner integrating EN 15804+A2 indicator individuals, lifecycle module value observations.
- **CEA v2.0** — developed by CEA List, structured around ESPR only as the regulatory alignment with rich semantics along the class/property.

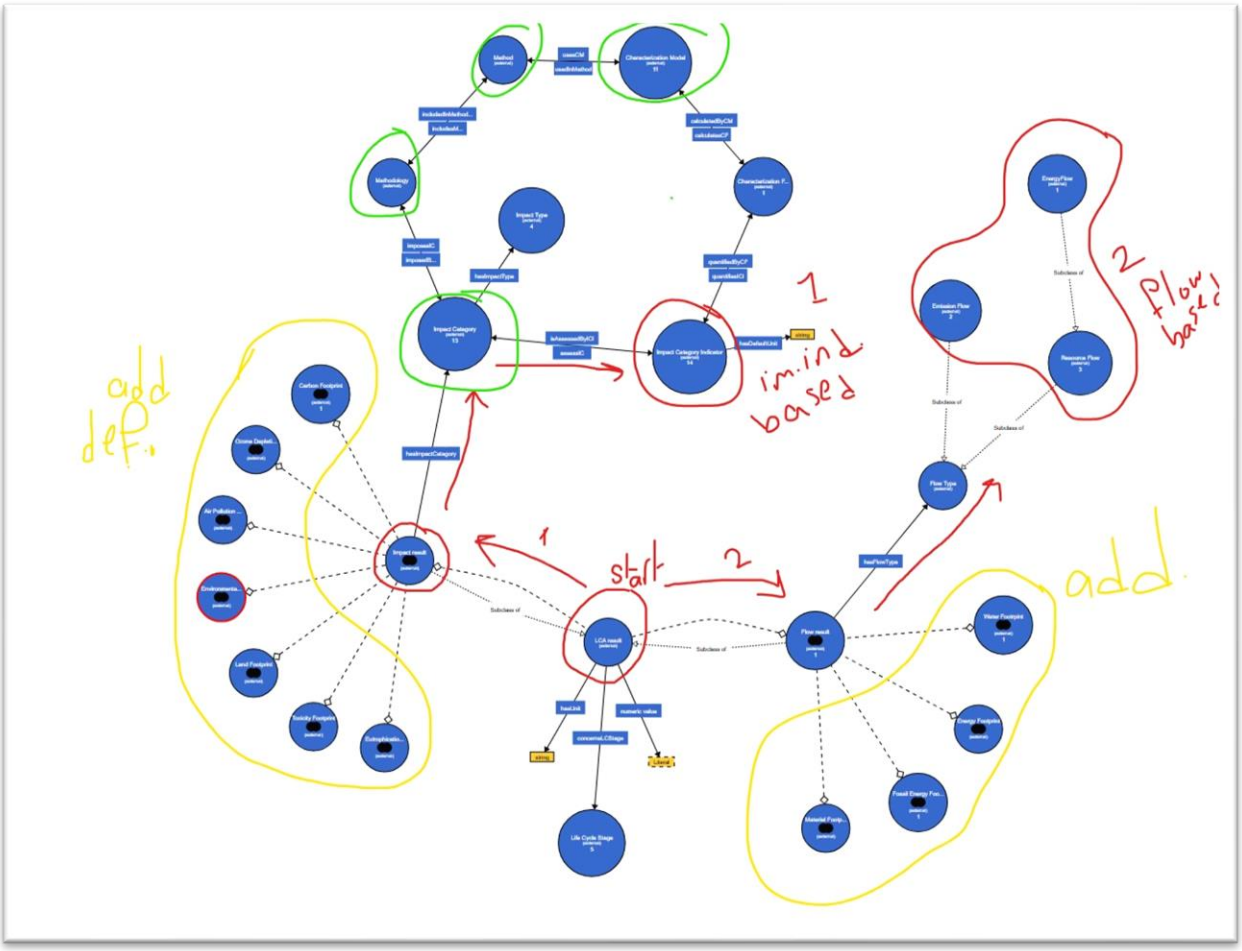
Following a comparative analysis of both ontologies — documented in the annotated diagrams attached to this specification (color-coded: red = structural similarity - a shared backbone, green = same concepts - modeled differently, yellow = additions of each,) — **Maki v2.0 was selected as the base for v3, with CEA's structural modifications to be incorporated within**. This decision rests on two grounds:

1. Maki's lean scope matches the DPP mandatory and non-mandatory regulations (ESPR, ILCD-base, ILCD+EPD extension) as well as following the EN15804+A2 and EF 3.1 controlled vocabulary of named individuals (19 EN 15804+A2 impact indicators, 18 inventory indicators, 16 EF 3.1 indicators) providing thus a **standards-grounded base**.
2. . CEA's structual formalization of classes/properties (e.g., indicator-centric result architecture (`LCAResult → quantifiesIndicator → LCIAIndicator`)) is semantically more correct than Maki's list-based declaration of classes/properties, having the ability thus to add **richer and cleaner semantics** to an extant base.

The changes and additions constituting v3 are documented in full in the sections below.



LCA_maki_v2.0.0



LCA_cea_v2.0.0

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2. Minor ontology Metadata Changes

2.1 Description Block

The following disclaimer block has been **removed** from the ontology's `dcterms:description`:

> *"DISCLAIMER: The scope is deliberately lean…" (full text removed — the scope statement is now implicit in the v3 structure itself and documented here instead of in the ontology header.)*

2.2 Namespace Declarations Added

The following namespace prefixes have been added to the ontology header:

```
xmlns:spdx="http://spdx.org/rdf/terms#"
xmlns:foaf="http://xmlns.com/foaf/0.1/"
xmlns:prov="http://www.w3.org/ns/prov#"
```

2.3 Versioning

- `owl:versionInfo`: "3.0.0"
- `owl:versionIRI`: https://w3id.org/eudpp/LCA/3.0.0
- `dcterms:creator`: Fatima Danash; Marc-Andree Wolf

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3. Minor class Hierarchy Changes

3.1 Renames

Previous name	v3 name	Reason
LCIAModuleValue	LCAModuleValue	Not specific to LCIA — also covers inventory module observations
LCIAIndicator	LCAIndicator	Same
LCIAMethdology	LCAMethodology	Same
LCIAOrInventoryMethod	LCA Method	Simpler; the qualifier was redundant given the ontology scope
EnvironmentalFootprintScore	EnvironemntalFootprintScoreResult	It is clearly a result

ALL comments have been respectively clarified.

3.2 Removed Classes

`Unit` class and `hasUnit` property — removed. Unit representation is already handled in the P_DPP module via the SI Digital Framework `MeasurementUnit` class and the `hasMeasurementUnit` object property (linking `QuantitativeProperty` to `MeasurementUnit`). A connector-level link between `LCAResult` and `QuantitativeProperty` will allow LCA results to inherit `hasMeasurementUnit` from P_DPP without redefining it here.

`ManufacturerRecord` class and `declaresManufacturerRecord` property — removed (fully redundant). The Actor module already covers:

- Manufacturer → `Actor` with `ManufacturerRole`
- Facility → `Facility` linked via `usesFacility`
- Location → `Location` linked via `isEstablishedIn` / `isResidingIn`

Cross-module action: objectProperty to be defined `:hasActor` (`LCAStudy` → `Actor`).

3.3 External Alignments Added

The following `rdfs:subClassOf` and `skos:broadMatch` alignments have been added to improve interoperability with standard vocabularies:

Class	Alignment	Target
Source	rdfs:subClassOf	prov:Entity
ComplianceDeclaration	rdfs:subClassOf	prov:Entity
EPDDocument	skos:broadMatch	foaf:Document
LCIAMethodReport	skos:broadMatch	foaf:Document
PCR	skos:broadMatch	foaf:Document
PEFCR	skos:broadMatch	foaf:Document
ReviewReport	skos:broadMatch	foaf:Document
LicenseType	rdfs:subClassOf	spdx:License
Review	rdfs:subClassOf	prov:Activity

Class	Alignment	Target
TypeOfReview	rdfs:subClassOf	skos:Concept
ComplianceStatus	rdfs:subClassOf	skos:Concept

4. Major class Hierarchy Additions

The following classes replace P_DPP placeholder classes, which will be removed from P_DPP following Tarmo's proposal. Cross-module `rdfs:subClassOf QuantitativeProperty` declarations for all these classes are deferred to the Connector module.

4.1 CarbonFootprint

A specialisation of `LCIAResult` representing the carbon footprint of a product system — the sum of greenhouse gas emissions and removals expressed in kg CO_2 equivalent, computed over a 100-year time horizon (ESPR Art. 2(25)). Instances quantify either the EN 15804+A2 GWP total indicator (`ind_GWPtotal`, an `EN15804ImpactIndicator`) or the EF 3.1 Climate Change indicator (`ind_EF_climateChange`, an `LCIAImpactIndicator`), depending on the reporting framework of the EPD or LCA study.

OWL restriction (SubClass Of):

> `( quantifiesIndicator value ind_GWPtotal ) OR ( quantifiesIndicator value ind_EF_climateChange )`

Connector module action: `CarbonFootprint rdfs:subClassOf QuantitativeProperty`

4.2 MaterialFootprint

A specialisation of `InventoryIndicatorResult` representing the material footprint of a product system — the total amount of raw materials extracted to meet final consumption demands (ESPR Art. 2(26)). Modelled as an inventory-level result (not a characterised LCIA result) because it aggregates raw material extraction flows without applying characterisation factors.

No `quantifiesIndicator` restriction is imposed, as material footprint aggregates across multiple raw material flow indicators in the EN 15804+A2 inventory (PERE, PENRE, SM, RSF, NRSF, etc.).

Connector module action: `MaterialFootprint rdfs:subClassOf QuantitativeProperty`

4.3 EnvironmentalFootprint (not added as a class)

ESPR Art. 2(24) defines environmental footprint as “a quantification of the environmental impacts resulting from a product throughout its life cycle, whether in relation to a single environmental impact category or an aggregated set of impact categories based on the Product Environmental Footprint method.”

This definition is intentionally broad — it covers both single-category results and fully aggregated scores. In the LCA module, `eudpp:LCAResult` already serves as the technical umbrella for all quantified LCA results. However, not every `LCAResult` constitutes an environmental footprint in the ESPR sense — `InventoryIndicatorResult` instances quantify inventory flows, not characterised impacts.

For this reason, `EnvironmentalFootprint` is **not introduced as a separate OWL class**. Instead, a `skos:broadMatch` annotation is added to `eudpp:LCAResult` pointing to the ESPR Art. 2(24) definition IRI, expressing that `LCAResult` broadly covers the regulatory concept without being formally equivalent to it.

The following sentence is added to the `rdfs:comment` of `eudpp:LCAResult`:

> *"In the regulatory sense of ESPR Art. 2(24), characterised subclasses of LCAResult (LCIAResult and EFSingleScoreResult) constitute environmental footprints — quantifications of environmental impacts of a product throughout its life cycle. InventoryIndicatorResult instances, however, quantify inventory flows rather than characterised impacts, and therefore fall outside the strict ESPR*

> *definition. For this reason, LCAResult is linked to the ESPR environmental footprint concept via skos:broadMatch rather than owl:equivalentClass."*

P_DPP action: `EnvironmentalFootprint` to be deleted from P_DPP.

4.4 EFSingleScoreResult

A specialisation of `LCAResult` representing the Environmental Footprint (EF) single score — a normalized and weighted aggregate of characterised results across all 16 EF 3.1 impact categories, expressed as a single dimensionless value (points, Pt). Computed by applying the normalization and weighting factors of the PEF method (Recommendation (EU) 2021/2279).

Modelled as a **sibling of `LCIAResult`** rather than a subclass of it, because the EF single score is qualitatively different from a per-category characterised result: it aggregates across all categories through normalization and weighting, steps that go beyond characterisation proper.

OWL restriction (SubClass Of):

> `quantifiesIndicator exactly 1 EnvironmentalFootprintScore`

Connector module action: `EFSingleScoreResult rdfs:subClassOf QuantitativeProperty`

4.5 ResourceConsumption Subclasses

Five new classes are added as **direct subclasses of `InventoryIndicatorResult`**, replacing the placeholder `ResourceConsumption` subclasses defined in P_DPP. No intermediate `ResourceConsumption` parent class is introduced.

Class IRI	Label	Typical EN 15804+A2 indicator(s)
`WaterConsumptionResult`	Water Consumption Result	FW (net fresh water)
`EnergyConsumptionResult`	Energy Consumption Result	PERE, PERM, PERT, PENRE, PENRM, PENRT
`LandUseResult`	Land Use Result	LU (note: classified as impact indicator in EF 3.1)
`RecycledMaterialsUseResult`	Recycled Materials Use Result	SM (secondary material)
`SustainableRenewableMaterialsUseResult`	Sustainable Renewable Materials Use Result	RSF, NRSF

Each class carries an `rdfs:label` matching its IRI local name and an `rdfs:comment` describing what it measures, noting the typical — though non-binding — correspondence to the EN 15804+A2 indicators listed above. No `quantifiesIndicator` restriction is imposed on any of these subclasses, to preserve flexibility for future indicator additions or revised characterisation frameworks. The `sourceVocabulary` annotation points to the LCA module IRI (`https://w3id.org/eudpp/LCA`).

Note on `LandUseResult`: its placement under `InventoryIndicatorResult` reflects alignment with the P_DPP `ResourceConsumption` grouping. Reviewers should note that LU is characterised as an impact indicator under EF 3.1 in EN 15804+A2; this placement may be revisited in a future version.

Connector module: All five classes declared as `rdfs:subClassOf QuantitativeProperty`.

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5. Major object Property Changes

5.1 Deleted Properties

`forLCIAOrInventoryMethod` — deleted. This property made the `Method` instance the primary link from a result, requiring a second hop through `forIndicator` to reach the indicator. This inverts the semantic priority: an LCA result is fundamentally a quantification of an indicator, not of a method. The method is computation metadata. EN 15804+A2 and ILCD+EPD both organise result reporting rows by indicator, not by method — confirming that the indicator is the primary identity anchor of a result.

Replaced by the object property `computedByMethod`, defined between `LCIAIndicator` and `Method`, with cardinality exactly 1 (`LCIAIndicator computedByMethod exactly 1 Method`). Backward compatibility is preserved via property chain axiom: `forLCIAOrInventoryMethod ≡ quantifiesIndicator ∘ inverse(computedByMethod)` if needed.

`forIndicator` — deleted. This property ran from `Method` to `LCIAIndicator`, making the method the subject and the indicator the object. In v3 the direction is inverted to `computedByMethod` (`LCIAIndicator → Method`), which correctly models the dependency: the indicator is the stable concept; the method is what computes it.

Replaced by object property `quantifiesIndicator` (`LCAResult → LCIAIndicator`), with cardinality exactly 1 per result instance.

`hasExternalDocumentation` — deleted (redundant with `referenceToDataSource`). Replaced by a unified property `hasExternalSource`, covering both (a) background documentation sources (project reports, safety data sheets, explanatory material) and (b) background databases or literature sources used for modelling (ecoinvent, GaBi/Sphera, Ökobaudat). The subproperty `hasReferenceToBackgroundDatabase` is retained as a specialisation for `Source` instances that are a `BackgroundDatabase`.

`declaresManufacturerRecord` — deleted

`hasUnit` — deleted

5.2 Renames

Previous name	v3 name
EN15804IndicatorGroup	belongsToEn15804Group
areaOfProtection	addressesAoP
referenceToIncludedMethods	includesMethod
complianceStatus	referenceToComplianceStatus
typeOfReview	hasTypeOfReview
Methodology	belongsToMethodology
referenceToDataSource	referenceToExternalSource
hasExternalDocumentation	hasExternalSource
licenseType	hasLicenseType
hasModuleValue	hasPerModuleValue
impactCategory	hasImpactCategory
referenceToExternaDocumentation(onLCIAorIventoryMethod)	referenceToExtenralSource
referenceToPCR	followsPCR
referenceToPEFCR	followsPEFCR
hasModuleValue	hasPerModuleValue

5.3 Enhanced semantics: Qualified Cardinality Restrictions on `quantifiesIndicator`

For `LCIAResult` and `InventoryIndicatorResult`, specialised range restrictions have been added via qualified cardinality axioms:

- **`LCIAResult`** is restricted to quantify exactly one indicator that is either an `EN15804ImpactIndicator` or an `LCIAImpactIndicator`. This captures the two study contexts in which a characterised impact result appears: an EPD study (EN 15804+A2 indicators) and a general LCA study (EF 3.1 / PEF indicators). The union allows both cases without forcing a choice at the class level.
- **`InventoryIndicatorResult`** is restricted to quantify exactly one `EN15804InventoryIndicator`. This reflects that inventory results are always linked to a defined inventory indicator, distinct from the characterised impact indicators used in `LCIAResult`.

The general range declaration (`quantifiesIndicator range LCIAIndicator`) on `LCAResult` is preserved and remains consistent with these restrictions, since `EN15804ImpactIndicator`, `LCIAImpactIndicator`, and `EN15804InventoryIndicator` are all subclasses of `LCIAIndicator`.

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6. Major object Property Additions

6.1 `relatesToLCIAIndicator`

Added between `EN15804ImpactIndicator` and `LCIAImpactIndicator`. Records a conceptual correspondence between the EN 15804+A2 reporting indicators and the EF 3.1 / PEF LCIA impact indicators, without implying equivalence or computational interchangeability. The property name “relates to” is deliberately the weakest possible semantic commitment — a conceptual link, not a formal mapping — because the two indicator sets serve distinct purposes and use different characterisation factor packages.

Assertions added (19 EN 15804 individuals → 16 EF 3.1 individuals):

EN 15804 individual	→	EF 3.1 / LCIA individual
ind_GWPtotal	→	ind_EF_climateChange
ind_GWPfossil	→	ind_EF_climateChange
ind_GWPbiogenic	→	ind_EF_climateChange
ind_GWPluluc	→	ind_EF_climateChange
ind_ODP	→	ind_EF_ozoneDepletion
ind_AP	→	ind_EF_acidification
ind_EPfreshwater	→	ind_EF_eutrophicationFreshwater
ind_EPmarine	→	ind_EF_eutrophicationMarine
ind_EPterrestrial	→	ind_EF_eutrophicationTerrestrial
ind_POCP	→	ind_EF_photochemicalOzoneFormation
ind_ADPE	→	ind_EF_resourceUseMineralsMetals
ind_ADPF	→	ind_EF_resourceUseFossils
ind_WDP	→	ind_EF_waterUse
ind_PM	→	ind_EF_particulateMatter
ind_IRP	→	ind_EF_ionisingRadiationHumanHealth

EN 15804 individual	→	EF 3.1 / LCIA individual
ind_ETPfw	→	ind_EF_ecotoxicityFreshwater
ind_HTPc	→	ind_EF_humanToxicityCancer
ind_HTPnc	→	ind_EF_humanToxicityNonCancer
ind_SQP	→	ind_EF_landUse

6.2 `assessesImpactCategory`

Added between LCIAImpactIndicator and LCIAImpactCategory. Records the classification link between each EF 3.1 impact indicator individual and the mid-point impact category it characterises. Domain is restricted to LCIAImpactIndicator only — not extended to EN15804ImpactIndicator — because EN 15804 impact indicators are already linked to their EF 3.1 counterparts via relatesToLCIAIndicator, from which the impact category is reachable transitively. Adding a direct assessesImpactCategory link from EN 15804 indicators would be redundant.

Assertions added (16 EF 3.1 individuals → 16 impact category individuals):

EF 3.1 individual	→	Impact category individual
ind_EF_climateChange	→	climateChange
ind_EF_acidification	→	acidification
ind_EF_eutrophicationFreshwater	→	eutrophication
ind_EF_eutrophicationMarine	→	eutrophication
ind_EF_eutrophicationTerrestrial	→	eutrophication
ind_EF_photochemicalOzoneFormation	→	photochemicalOzoneFormation
ind_EF_ozoneDepletion	→	ozoneDepletion
ind_EF_ionisingRadiationHumanHealth	→	ionisingRadiationHumanHealth
ind_EF_resourceUseFossils	→	resourceUseFossil
ind_EF_ecotoxicityFreshwater	→	freshwaterEcotoxicity
ind_EF_humanToxicityCancer	→	humanToxicity
ind_EF_humanToxicityNonCancer	→	humanToxicity
ind_EF_landUse	→	landUseSoilQuality
ind_EF_resourceUseMineralsMetals	→	resourceUseMineralsMetals
ind_EF_waterUse	→	waterUseDeprivation
ind_EF_particulateMatter	→	respiratoryInorganics

6.3 `addressesAoP` — Individual Assertions

Individual assertions added between LCIAImpactCategory individuals and AreaOfProtection individuals, based on EF 3.1 (Andreasi Bassi et al., JRC, 2023):

Impact category individual	→	Area(s) of protection
climateChange	→	aopHumanHealth, aopNaturalEnvironment

Impact category individual	→	Area(s) of protection
ozoneDepletion	→	aopHumanHealth, aopNaturalEnvironment
humanToxicity	→	aopHumanHealth
respiratoryInorganics	→	aopHumanHealth
ionisingRadiationHumanHealth	→	aopHumanHealth
photochemicalOzoneFormation	→	aopHumanHealth, aopNaturalEnvironment
acidification	→	aopNaturalEnvironment
eutrophication	→	aopNaturalEnvironment
freshwaterEcotoxicity	→	aopNaturalEnvironment
landUseSoilQuality	→	aopNaturalEnvironment, aopNaturalResources
waterUseDeprivation	→	aopHumanHealth, aopNaturalEnvironment, aopNaturalResources
resourceUseFossil	→	aopNaturalResources
resourceUseMineralsMetals	→	aopNaturalResources
terrestrialEcotoxicity	→	aopNaturalEnvironment
marineEcotoxicity	→	aopNaturalEnvironment
ionisingRadiationEcosystems	→	aopNaturalEnvironment

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7. Major data Property Changes

7.1 `amount` — Domain Extension

Domain widened from `LCAModuleValue` alone to `LCAModuleValue ∪ LCAResult` (union domain). This allows an `LCAResult` instance to carry a direct numeric value for aggregated or non-module-specific declarations, while `LCAModuleValue` continues to carry module-specific values (e.g. per lifecycle module A1–A3, C3, D). The two usages are not mutually exclusive: a result can have both a direct `amount` and one or more `LCAModuleValue` instances with their own `amount`.

Updated `rdfs:comment`:

> *"Numerical value of an LCA result. When declared on an LCAModuleValue instance, it represents the value for a specific life cycle module (e.g. A1–A3, C3, D) in the unit given by the associated LCIA method. When declared directly on an LCAResult instance, it represents an aggregated or non-module-specific value."*

7.2 Standard Vocabulary Alignments (subPropertyOf)

The following data properties are declared as subproperties of standard Dublin Core terms:

- `dataSetVersion`, `EPDFormatVersion`, `ILCDFormatVersion`, `methodVersion`, `methodologyVersion`, `modelVersion` → `rdfs:subPropertyOf dcterms:hasVersion`
- `registrationNumber`, `UUID` → `rdfs:subPropertyOf dcterms:identifier`

7.3 Replaced by Standard Vocabulary

The following local data properties have been deleted and replaced by direct use of standard terms:

- `impactIndicatorDescription` → `dcterms:description`

- `baseName` → `rdfs:label`
- `modelName` → `rdfs:label`

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8. Major annotation Property Change

8.1 `ilcdSource` → `sourceVocabulary`

The annotation property `ilcdSource` has been renamed to `sourceVocabulary`. The original name framed the ontology as derivative of the ILCD data format, implying that ILCD was the reference point against which all elements were measured. This framing is incorrect: this ontology is an LCA and EPD ontology designed for DPP purposes, which aligns with external vocabularies and standards — including ILCD — but does not originate from them.

Updated values:

Value	Status	Meaning
"LCA ontology"	New (merges previous two)	Native element; no direct external counterpart. Replaces the previous values "LCA ontology extension" and "LCA ontology metadata", which have been merged.
"ILCD base"	Retained	Originates from the ILCD 1.1 base data format
"ILCD+EPD extension"	Retained	Originates from the ILCD+EPD v1.3 extension
"Dublin Core"	New	Aligned with Dublin Core Terms
"PROV-O"	New	Aligned with W3C PROV-O
"SKOS"	New	Aligned with W3C SKOS
"FOAF"	New	Aligned with FOAF vocabulary
"LCAnet"	New	Aligned with LCAnet ontology

All `rdfs:comment` values and `sourceVocabulary` annotations across the ontology have been updated accordingly.

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9. Major named Individual Changes

All EN 15804 lifecycle module individuals shared IRIs with COMP conformity assessment module individuals (`moduleA1`, …, `moduleD`), causing a one-IRI–two-types conflict when both modules are loaded in the same graph. These have been renamed with the `en15804` prefix.

Other individuals of ImpactCategory also shared the same IRI with individuals of LCAIndicators. These were corrected.

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10. Major Connector and P_DPP Module Actions

The following cross-module declarations are **not part of this LCA module file** and must be added in the Connector module:

1. `LCAResult rdfs:subClassOf QuantitativeProperty` (and `LCAModuleValue rdfs:subClassOf QuantitativeProperty`) — to allow LCA results to inherit `hasMeasurementUnit` from P_DPP.

Note that the LCAResult class has the following subclasses:



Accordingly, they will all be subclasses of `QuantitativeProperty`.

2. `hasActor` object property (`LCAStudy → Actor`).

P_DPP action required (based on Tarmo's proposal because these elements now exist in LCA module)

1. `EnvironmentalFootprint` class and its subclasses (MaterialFootprint and CarbonFootprint) to be deleted from P_DPP.
2. `ResourceConsumption` class and its subclasses (LandUse, SustainableRenewalMaterialsUse, REcycledMaterialsUse, WaterConsumption, and EnergyConsumption) to be deleted from P_DPP